Inhibitory effect of marinades with hibiscus extract on formation of heterocyclic aromatic amines and sensory quality of fried beef patties.

Heterocyclic aromatic amines (HAA) are carcinogenic compounds found in the crust of fried meat. The objective was to examine the possibility of inhibiting HAA formation in fried beef patties by using marinades with different concentrations of hibiscus extract (Hibiscus sabdariffa) (0.2, 0.4, 0.6, 0.8 g/100g). After frying, patties were analyzed for 15 different HAA by HPLC-analysis. Four HAA MeIQx (0.3-0.6 ng/g), PhIP (0.02-0.06 ng/g), co-mutagenic norharmane (0.4-0.7 ng/g), and harmane (0.8-1.1 ng/g) were found at low levels. The concentration of MeIQx was reduced by about 50% and 40% by applying marinades containing the highest amount of extract compared to sunflower oil and control marinade, respectively. The antioxidant capacity (TEAC-Assay/Folin-Ciocalteu-Assay) was determined as 0.9, 1.7, 2.6 and 3.5 micromol Trolox antioxidant equivalents and total phenolic compounds were 49, 97, 146 and 195 microg/g marinade. In sensory ranking tests, marinated and fried patties were not significantly different ($p>0.05$) to control samples.